

A ZFC Overcomplete Set in $C([0, \omega_1])$ with No Weak Cluster Point

Verification and problem exploration report

June 22, 2026

Abstract

Russo and Somaglia asked whether $C([0, \omega_1])$ contains an overcomplete set with no weak cluster point, and whether (for example under CH) an overcomplete set can be weakly discrete. The supplied partial packet gives an affirmative answer under CH. We verify that argument, apart from a minor net-selection detail that is easily repaired.

A literature search located Koszmider's later ZFC construction of an overcomplete set in $C([0, \omega_1])$, but no published resolution of the additional weak-topological requirement. We then give a ZFC construction which appears to settle both questions affirmatively. In fact, for every $0 < \varepsilon < 1/2$ we construct an overcomplete family $(x_\alpha)_{\alpha < \omega_1}$ satisfying

$$\|x_\alpha - \mathbf{1}_{[0, \alpha]}\| < \varepsilon \quad (\alpha < \omega_1).$$

It follows that the family is norm separated and weakly closed-discrete. The argument combines the almost coherent injections used in Koszmider's nonseparable version of Klee's construction with a small analytic term whose Taylor coefficients grow unboundedly. This report is not peer reviewed, so the proposed ZFC resolution should be independently checked before being cited as established literature.

Packet status

Run `fa_banach_001`.

Source arXiv id
`1912.08690`.

Source paper T. Russo and J. Somaglia, *Overcomplete sets in non-separable Banach spaces*, arXiv:1912.08690.

Packet classification

Candidate full solution, likely valid subject to human mathematical review.

Upgrade note

This packet supersedes the earlier CH partial packet for the same source question. The earlier packet is preserved in `history/packets/`; the new theorem removes CH and gives a ZFC weakly closed-discrete overcomplete set in $C([0, \omega_1])$.

The source question appears in Section 4.1 of Russo–Somaglia:

the transfinite induction and, thereby, the proof. $\square$

One might wonder if the above result can be improved in order to produce overcomplete sets with no cluster point also in the weak topology. It is, however, elementary to realise that this is, in general, not possible. Indeed, every WLD Banach space is Lindelöf in the weak topology (see, *e.g.*, [17, Theorem 17.1]) and it is clear that every uncountable subset of a Lindelöf topological space needs to have a cluster point. On the other hand, we do not know what happens for Banach spaces that are not WLD. For example, is it possible to find an overcomplete set in $C([0, \omega_1])$ with no weak cluster point? Moreover, we do not know if it is possible, say under CH, to construct overcomplete sets that are discrete in the weak topology.

1 The question and the status of the supplied partial result

For a Banach space X , a set $S \subseteq X$ is *overcomplete* if $|S| = \text{dens } X$ and every subset of S having cardinality $|S|$ is linearly dense in X . Russo and Somaglia asked whether one can find an overcomplete set in $C([0, \omega_1])$ with no weak cluster point. They also asked whether, say under CH, one can construct overcomplete sets which are discrete for the weak topology [1, Section 4.1].

The supplied packet proposes the following CH argument. Put $u_\alpha = \mathbf{1}_{[0, \alpha]}$. Small norm balls around the u_α 's are locally finite in the topology of pointwise convergence. Under CH, enumerate all hyperplanes of $C([0, \omega_1])$ in order type ω_1 and choose, at stage α , a point close to u_α and outside all preceding hyperplanes. Every hyperplane then contains only countably many selected points, which is exactly the hyperplane criterion for overcompleteness.

Proposition 1 (Audit of the supplied packet). *The CH theorem in the supplied packet is correct. Its local-finiteness argument needs only the following minor clarification: failure of local finiteness gives a net of indices which eventually avoids every finite subset of ω_1 , rather than automatically a globally injective net. That is all the proof needs. Moreover, the radius restriction can be improved from $r < 1/3$ to $r < 1/2$.*

Proof. The repaired local-finiteness proof is given in Lemma 2 below. Once that lemma is available, the CH recursion is standard. At every countable stage, a nonempty norm-open ball cannot be covered by countably many hyperplanes and points, by the Baire category theorem. If an uncountable subfamily failed to be linearly dense, a nonzero functional would annihilate it, so its kernel would be one of the enumerated hyperplanes and would contain uncountably many selected points, contradicting the recursion. Local finiteness of the containing balls then makes the selected set weakly closed-discrete. $\square$

2 Literature search

The source problem appears in the 2021 paper of Russo and Somaglia [1]. Koszmider subsequently proved in ZFC that $C([0, \omega_1])$ admits an overcomplete set [2, Theorem 11]; his argument uses a coherent family of injections and a nonseparable analogue of Klee's analytic construction. That theorem settles plain existence, but its statement and proof do not impose the absence of weak cluster points. The 2022 hyperplane-covering follow-up [3] addresses a different overcomplete-set problem. Russo's 2025 habilitation overview still summarizes this part of the theory through the original paper and the two follow-ups, without recording a resolution of the weak-topological question [4, pp. 16–17].

Exact-phrase searches for the displayed question, “weak cluster point”, “weakly discrete”, and “weakly closed-discrete”, together with searches of the citation trail and current author publication lists, did not locate a published proof or counterexample through June 22, 2026. This is necessarily a bounded novelty check, not a proof that no such result exists.

3 Proof Intuition

The old CH argument used a well ordering of all hyperplanes and chose points near the clopen initial-segment functions $u_\alpha = \mathbf{1}_{[0,\alpha]}$. The ZFC construction replaces that global enumeration with a direct functional-separation mechanism.

Every functional on $C([0, \omega_1])$ has countable support, so its cumulative values $\mu(u_\alpha)$ are eventually constant. The terms $u_\beta - u_\alpha$ cancel this eventual constant. Almost coherent injections then make the values of a fixed functional on an uncountable subfamily into zeros of one analytic function. The additional scalar term $q(r_\alpha)\mathbf{1}_K$, with $q(t) = t/(1 - 2t)$, detects the remaining endpoint mass at ω_1 : its Taylor coefficients grow like 2^n , while the coefficients coming from a bounded functional stay bounded. Therefore a nonzero functional cannot vanish on uncountably many constructed vectors.

The weak-topological control is kept by forcing $\|x_\alpha - u_\alpha\| < \varepsilon < 1/2$. Small balls around the u_α ’s are weakly locally finite, so the final overcomplete set has no weak cluster point and is weakly closed-discrete.

4 Two preliminary lemmas

Let

$$K = [0, \omega_1], \quad X = C(K), \quad u_\alpha = \mathbf{1}_{[0,\alpha]} \quad (\alpha < \omega_1).$$

Each $[0, \alpha]$ is clopen, so $u_\alpha \in X$. We first record the weak-topological part of the supplied argument in a fully explicit form.

Lemma 2. *For every $0 < r < 1/2$, the family*

$$\mathcal{B}_r = \{B(u_\alpha, r) : \alpha < \omega_1\}$$

is locally finite for the topology of pointwise convergence on X . Consequently it is locally finite for the weak topology of X .

Proof. Suppose that $\mathcal{B}_r$ is not locally finite at some $f \in X$. Direct the pairs (V, F) , where V is a pointwise neighbourhood of f and $F \subseteq \omega_1$ is finite, by refinement of V and enlargement of F . For every such pair choose

$$\alpha_{V,F} \notin F, \quad g_{V,F} \in V \cap B(u_{\alpha_{V,F}}, r).$$

Then $g_{V,F} \rightarrow f$ pointwise, while the index net $(\alpha_{V,F})$ eventually avoids every finite subset of ω_1 .

By compactness of K , pass to a subnet for which $\alpha_{V,F} \rightarrow \lambda \in K$. The subnet is not eventually constant. Since all isolated points of the ordinal compactum force eventually constant convergence, λ is a limit ordinal, possibly ω_1 . A direct check in the order topology gives

$$u_{\alpha_{V,F}}(t) \longrightarrow v_\lambda(t) \quad (t \in K), \quad v_\lambda = \mathbf{1}_{[0,\lambda]}.$$

Indeed, for $t < \lambda$ the indices are eventually above t , whereas at $t = \lambda$ they are eventually strictly below λ ; for $t > \lambda$ they are eventually below t .

For every $t \in K$ we therefore have

$$|f(t) - v_\lambda(t)| \leq r.$$

In particular,

$$|f(\lambda)| \leq r, \quad |f(\xi) - 1| \leq r \quad (\xi < \lambda),$$

so

$$|f(\xi) - f(\lambda)| \geq 1 - 2r > 0 \quad (\xi < \lambda).$$

This contradicts continuity of f at the limit point λ . Hence the family is pointwise locally finite. Since every coordinate evaluation δ_t is weakly continuous, every pointwise neighbourhood used above is also weakly open, and weak local finiteness follows. $\square$

The combinatorial device below is standard and is used explicitly in Koszmider's construction [2, Lemma 9]. We include a proof for completeness. For two functions on a common domain, $p =^* q$ means that they differ at only finitely many points.

Lemma 3 (Almost coherent injections). *There are injections $e_\alpha : \alpha \rightarrow \mathbb{N}$ for $\alpha < \omega_1$ such that*

$$e_\beta \upharpoonright \alpha =^* e_\alpha \quad (\alpha < \beta < \omega_1).$$

They may moreover be chosen so that $\mathbb{N} \setminus e_\alpha[\alpha]$ is infinite for every α .

Proof. Proceed by transfinite recursion. At a successor stage, extend e_α by sending the new point to an unused integer; the infinite-complement condition is preserved.

Let $\lambda < \omega_1$ be a limit ordinal, and choose an increasing cofinal sequence $(\lambda_n)_{n < \omega}$ in λ . Recursively construct injections $h_n : \lambda_n \rightarrow \mathbb{N}$ so that h_{n+1} extends h_n and $h_n =^* e_{\lambda_n}$. To pass from n to $n+1$, use $e_{\lambda_{n+1}}$ on the new block $\lambda_{n+1} \setminus \lambda_n$. Only finitely many of those values collide with $h_n[\lambda_n]$, because $h_n =^* e_{\lambda_{n+1}} \upharpoonright \lambda_n$; alter those finitely many values to fresh integers. There are infinitely many available integers, since the union in question is a finite perturbation of $e_{\lambda_{n+1}}[\lambda_{n+1}]$.

Set $h = \bigcup_n h_n$. Then h is injective and $h \upharpoonright \lambda_n =^* e_{\lambda_n}$ for every n . If the complement of $h[\lambda]$ is not infinite, choose a countable set $B = \{\xi_n : n < \omega\} \subseteq \lambda$ such that $B \cap \lambda_m$ is finite for every m (take $\xi_n \geq \lambda_n$), partition B into infinitely many infinite sequences, and on each sequence shift the h -values forward by one place. This preserves injectivity, changes h only finitely often on every λ_m , and frees one value from each sequence. The resulting injection is e_λ . Coherence with every earlier ordinal follows from cofinality of (λ_n) and transitivity of $=^*$. $\square$

Corollary 4 (Exact stabilization on an initial segment). *Let (e_α) be as in Lemma 3. If $A \subseteq \omega_1$ is uncountable and $\gamma < \omega_1$, then there are an uncountable $A' \subseteq A \setminus (\gamma + 1)$ and an injection $g : \gamma + 1 \rightarrow \mathbb{N}$ such that*

$$e_\alpha \upharpoonright (\gamma + 1) = g \quad (\alpha \in A').$$

Proof. For $\alpha > \gamma$, the restriction $e_\alpha \upharpoonright (\gamma + 1)$ differs from $e_{\gamma+1}$ on a finite set. Since $\gamma + 1$ is countable, there are only countably many finite subsets of it; thin A to an uncountable set on which that exceptional set is fixed. There are only countably many maps from the fixed finite exceptional set into $\mathbb{N}$, so a second thinning makes the entire restriction constant. $\square$

5 The ZFC construction

The key design is as follows. For a functional μ , the cumulative values $\mu(u_\alpha)$ are eventually constant, because μ has countable support. The differences $u_\beta - u_\alpha$ cancel that eventual constant. The coherent injections then turn evaluation on an uncountable subfamily into the zero set of one analytic function. Those differences all vanish at ω_1 , however, so an extra constant-vector term is needed to detect the atom at ω_1 . We choose that term to have unbounded Taylor coefficients, whereas all coefficients coming from a bounded functional remain uniformly bounded.

Theorem 5 (ZFC). *For every $0 < \varepsilon < 1/2$, there is a family $S = \{x_\alpha : \alpha < \omega_1\} \subseteq C([0, \omega_1])$ such that*

- (i) $\|x_\alpha - u_\alpha\| < \varepsilon$ for every $\alpha < \omega_1$;
- (ii) for every nonzero $\mu \in C([0, \omega_1])^*$, the set $\{\alpha < \omega_1 : \mu(x_\alpha) = 0\}$ is countable.

Consequently S is overcomplete, bounded, $(1 - 2\varepsilon)$ -separated in norm, and weakly closed-discrete. In particular, S has no weak cluster point.

Proof. Fix $b \in (0, 1/2)$ so small that

$$\frac{b}{1-b} + \frac{b}{1-2b} < \varepsilon. \quad (1)$$

Choose pairwise distinct numbers $r_\alpha \in [b/2, b]$ for $\alpha < \omega_1$; this is possible because every nondegenerate real interval has cardinality $\mathfrak{c} \geq \omega_1$. Let $(e_\alpha)_{\alpha < \omega_1}$ be as in Lemma 3, and put

$$q(t) = \frac{t}{1-2t} = \sum_{n=0}^{\infty} 2^n t^{n+1} \quad (|t| < 1/2).$$

For $\alpha < \omega_1$, define

$$w_\alpha = \sum_{\beta < \alpha} r_\alpha^{e_\alpha(\beta)+1} (u_\beta - u_\alpha), \quad (2)$$

$$x_\alpha = u_\alpha + w_\alpha + q(r_\alpha) \mathbf{1}_K. \quad (3)$$

The series in (2) converges absolutely in norm, since e_α is injective and $\|u_\beta - u_\alpha\| = 1$ for $\beta < \alpha$. Moreover,

$$\|w_\alpha\| \leq \sum_{n=0}^{\infty} r_\alpha^{n+1} = \frac{r_\alpha}{1-r_\alpha} \leq \frac{b}{1-b}.$$

Together with (1), this gives

$$\|x_\alpha - u_\alpha\| \leq \frac{b}{1-b} + \frac{b}{1-2b} < \varepsilon,$$

which proves (i).

We prove (ii). Since K is scattered, every functional on $C(K)$ is an atomic Radon measure with countable support; equivalently, $C(K)^* \cong \ell_1(K)$ [5]. Fix $\mu \in C(K)^*$ and write

$$a_\xi = \mu(u_\xi) \quad (\xi < \omega_1), \quad d = \mu(\mathbf{1}_K).$$

The support of μ below ω_1 is countable and therefore bounded in ω_1 . Hence there are $\gamma < \omega_1$ and $c \in \mathbb{R}$ such that

$$a_\xi = c \quad (\xi > \gamma). \quad (4)$$

Suppose that $A = \{\alpha : \mu(x_\alpha) = 0\}$ is uncountable. Discarding an initial segment and applying Corollary 4, find an uncountable $A' \subseteq A \setminus (\gamma + 1)$ and an injection $g : \gamma + 1 \rightarrow \mathbb{N}$ such that $e_\alpha \upharpoonright (\gamma + 1) = g$ for every $\alpha \in A'$.

For $\alpha \in A'$, equations (2) and (4) give

$$\mu(w_\alpha) = \sum_{\beta \leq \gamma} (a_\beta - c) r_\alpha^{g(\beta)+1}.$$

Thus the analytic function

$$H(t) = c + \sum_{\beta \leq \gamma} (a_\beta - c) t^{g(\beta)+1} + d q(t) \quad (5)$$

vanishes at every r_α with $\alpha \in A'$. The middle series is analytic on the unit disc because its coefficients are bounded and g is injective; q is analytic on $|t| < 1/2$. The uncountable subset $\{r_\alpha : \alpha \in A'\}$ of the compact interval $[b/2, b]$ has an accumulation point inside $(-1/2, 1/2)$. The same power series defines a holomorphic function on the complex disc $|t| < 1/2$, so the identity theorem yields $H \equiv 0$ there.

The constant coefficient in (5) is c , so $c = 0$. For $n < \omega$, let

$$B_n = \begin{cases} a_\beta, & n = g(\beta) \text{ for some } \beta \leq \gamma, \\ 0, & n \notin g[\gamma + 1]. \end{cases}$$

Comparing the coefficient of t^{n+1} in $H \equiv 0$ gives

$$B_n + d 2^n = 0 \quad (n < \omega).$$

The sequence (B_n) is bounded, since $|a_\beta| \leq \|\mu\|$. Hence the last equalities force $d = 0$, and then $B_n = 0$ for every n . Consequently $a_\beta = 0$ for $\beta \leq \gamma$, while (4) and $c = 0$ give $a_\xi = 0$ for all $\xi > \gamma$. Therefore

$$\mu(\mathbf{1}_K) = 0, \quad \mu(u_\xi) = 0 \quad (\xi < \omega_1).$$

The linear span of $\mathbf{1}_K$ and the u_ξ is a unital algebra which separates points of K (indeed $u_\xi(\xi) = 1$ and $u_\xi(\eta) = 0$ for $\xi < \eta$). By the Stone–Weierstrass theorem it is dense in $C(K)$, so $\mu = 0$. Thus a nonzero functional cannot vanish on uncountably many x_α 's, proving (ii).

We finish with the stated consequences. The step-function algebra above has a dense subset of cardinality ω_1 , while $(u_\alpha)_{\alpha < \omega_1}$ is 1-separated; hence $\text{dens } C(K) = \omega_1$. Since $\|u_\alpha - u_\beta\| = 1$ for $\alpha \neq \beta$, condition (i) implies

$$\|x_\alpha - x_\beta\| > 1 - 2\varepsilon,$$

so S has cardinality ω_1 and is norm separated. Also $\|x_\alpha\| < 1 + \varepsilon$, so S is bounded. If an uncountable subfamily were not linearly dense, Hahn–Banach would provide a nonzero functional annihilating it, contradicting (ii). Hence S is overcomplete. Finally, $x_\alpha \in B(u_\alpha, \varepsilon)$ and the family of these balls is weakly locally finite by Lemma 2. Every point of X has a weak neighbourhood meeting only finitely many members of S ; by the Hausdorff property, one can remove those finitely many points except possibly the centre itself. Thus S is weakly closed-discrete. $\square$

6 Consequences and comparison with the partial packet

Theorem 5 answers both questions from [1, Section 4.1] in ZFC, and gives substantially more than the CH packet:

- no enumeration of all hyperplanes is used;
- the set can be chosen uniformly as close as desired to the clopen step family (u_α) ;
- every nonzero functional vanishes on only countably many selected vectors, an explicit strengthening of the hyperplane criterion;
- the resulting set is bounded, norm separated, and weakly closed-discrete.

The construction can be viewed as a localization of Koszmider’s ZFC analytic method. The term $u_\beta - u_\alpha$ removes the eventual constant in the cumulative values of a countably supported measure. The function $q(t) = t/(1 - 2t)$ encodes the remaining one-dimensional endpoint obstruction: its coefficients 2^n grow without bound, while coefficients obtained by evaluating norm-one step functions under a fixed bounded functional remain bounded. This is the point that upgrades “codimension at most one” to full overcompleteness while retaining the prescribed weakly discrete geometry.

Verification and novelty caveat

The proof has been checked line by line against the following possible failure points: convergence of the transfinite series, exact stabilization of the coherent injections, the eventual constancy of cumulative masses, location of an interior accumulation point for the analytic identity theorem, boundedness of the coefficients B_n , density of the step-function algebra, and the passage from local finiteness to weak closed-discreteness. No contradiction was found. Nevertheless, the ZFC theorem in this report was produced during this exploration and has not undergone external peer review. The literature search did not reveal an existing answer, so independent expert verification is particularly important.

References

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